

and enjoys extensive developer support.

Ogg Vorbis is a lossy codec that compresses music in a technique similar to, but vastly better than MP3. It supports VBR, which lets you tweak the settings to get fine fidelity for less space. There is no encoding quality limitation specified—the encoders can support an amazing 16 to 500 Kbps in stereo and 32 to 256 Kbps in mono mode. Here, quality is not measured in kilobits per second, instead an arbitrary 10-point scale is used—quality level 0 is equivalent to 64 Kbps, level 5 is roughly 160 Kbps and level 10 is about 400 Kbps. If you want near Audio CD quality, stick to level 3 or 4 in which the audio quality and file size are best adjusted.

Most players have been supporting Ogg Vorbis through plugins for some time now. Winamp, for instance, natively supports Ogg Vorbis. On the encoding front, software such as dBPowerAMP converts your existing audio files to Ogg Vorbis within a few clicks. The software support is expected to significantly increase over time, with more and more users seeing Ogg Vorbis as the format that offers the right mix of quality and size. The music industry is veering sharply away though—Ogg Vorbis, like MP3 and unlike WMA, has no safeguards against piracy.

Hardware support too is sadly lacking—you will have to wait a while to play Vorbis files on your portable boom box. Iomega and Rio are two companies all set to make Vorbis playing devices, if the consumer demand is strong enough to support them.

Ogg Vorbis is a potential rival to the MP3 format and it will continue to get better because of the flexibility that allows for significant tuning and training of the algorithm, even after the format is frozen—and all this for free, with no copyrights and patents!

AAC

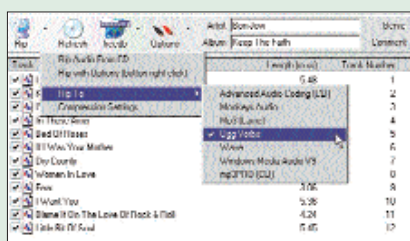
Web site: <http://www.vialicensing.com/products/mpeg4aac/standard.html>

One of the most promising formats on the skyline is the patented AAC (Advanced Audio Coding) standard. In the late 1990s, this format was sidelined by MP3, as it demanded unrealistic computing resources for encoding and decoding. But now, AAC is being promoted as the audio data compression standard of the 21st century. It is now an official standard under ISO-MPEG, with Via Licensing currently in charge of licensing the technology.

AAC has become the base for a num-

Start the Ripping!

The best part about working with audio formats is that most of the shareware and free jukebox software can mix a lot of compressed audio formats. We will use the dBPowerAMP Music Converter to convert pure uncompressed audio into high-quality Ogg Vorbis, WMA and AAC files. It has a simple interface and offers the convenience of single-click conversions from within Windows Explorer. It also supports a wide range of lossy and lossless formats by way of plugins. For MP3PRO enthusiasts, try the demo player/encoder by Thomson Multimedia



dBPoweramp Music Converter, a one-stop-solution for most format conversions

that plays both MP3 and MP3PRO files.

■ With the Audio CD of choice in the CD-ROM drive, go to **Start > Programs > dBPower-AMP Music Converter** and click on dBPower-AMP Music Converter Audio CD Input. This will open the CD dialog box, listing out the Audio CD track details. Choose **Rip > Rip To** and choose the format that you want to encode your music to.

■ Once you set your format, all you have to do is click Rip, and encode with the preset values. If you want to tinker under the hood and play with the quality settings, then try **Rip > Rip with options**, or simply right-click on Rip. This will open up the codec options

box. Click on Convert to start encoding.

If you already have wave files or compressed audio files that you would like to encode into a different format, just navigate to the file's location in Explorer, right-click and choose Convert To. Then select the format of your choice and access the options. After that, click on Convert to start the process.



Thomson MP3PRO Audio Player encodes your MP3PROs

For MP3PRO, you can use the demo MP3PRO player/encoder by clicking on Select to choose the source file. Next, click on Encode to access the encoding facility. Click Encode again on the dialog box to generate the MP3PRO file. For all formats, experiment with VBR for better compression quality.

You can also use Nero Burning Rom, however, the bundled plugin is a demo version that allows only 30 operations. Its encoding quality ranges from 24 Kbps for the mono mode to 96 Kbps for the stereo mode.

Avoid converting your existing MP3 files into any of these formats—instead of the superior audio quality you expect, there will be further loss in quality. Instead, re-rip from your Audio CDs and encode into these formats to get clear, high fidelity audio.

ber of sophisticated audio codecs, including AT&T's a2b and will also be used in the MPEG-4 standard. The standard can record data in both mono and stereo mode, and up to a maximum of 48 channels of data! Add to that sampling rates of 96 KHz and broadcast quality at a bit rate of 320 Kbps for 5.1-channel sound and you have the future of compressed digital audio.

The standard is made up of three different 'profiles', which differ from each other in sound quality. Studies conducted by MPEG show that AAC profiles encoded at 128 Kbps and 96 Kbps bit rates rank way ahead of MP2 at 192 Kbps and MP3 at 128 Kbps. While this and numerous

other tests show that AAC ranks ahead of the contenders, including WMA, its popularity is severely restricted by the fact that it is very tightly licensed. This may change though—AAC is widely regarded as one of the highest-quality formats for distributing music on the Internet and is also targeted towards streaming audio. However, despite improvements in computing technology, the processor requirements for both encoding and decoding are rather high. Moreover, software support is quite limited—as far as decoders go, there is a plugin for Winamp. On the encoder side, AACenc is available for non-commercial use only, while you can